

Chittagong University of Engineering & Technology



Department of Electrical & Electronic Engineering

Course No.: EEE 460

Course Title: Electrical Service Design

Project Title:

**Automatic Power Factor Corrector System & Electrical
Design for Tier-3 Data Center**

Project Report

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1 Project Objectives

In a data center environment, the proliferation of non-linear loads (servers, UPS systems, variable frequency drives for cooling) creates significant reactive power demand and harmonic distortion. The objectives of our APFC design are:

- **Efficiency Optimization:** To maintain the Power Factor (PF) close to unity (0.99), thereby reducing transmission losses and ensuring compliance with utility grid regulations (DESCO/BPDB).
- **Harmonic Mitigation:** Utilizing Detuned Reactors (7% impedance) to filter out harmful harmonics that can cause overheating in transformers and server power supplies.
- **Cost Reduction:** To minimize kVA demand charges and avoid low power factor penalties imposed by utility providers.
- **Voltage Stability:** To improve voltage regulation at the main busbar, ensuring stable operation of sensitive IT equipment during load fluctuations.

2 Layout Designs

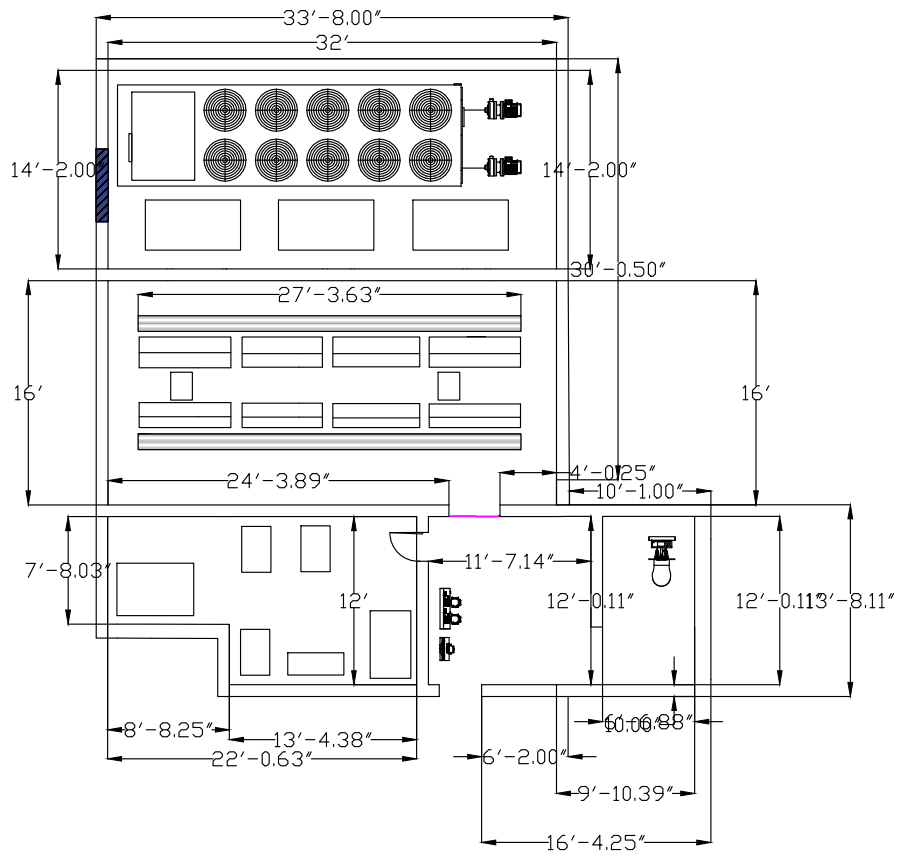


Figure 1: Layout Measurement Plan

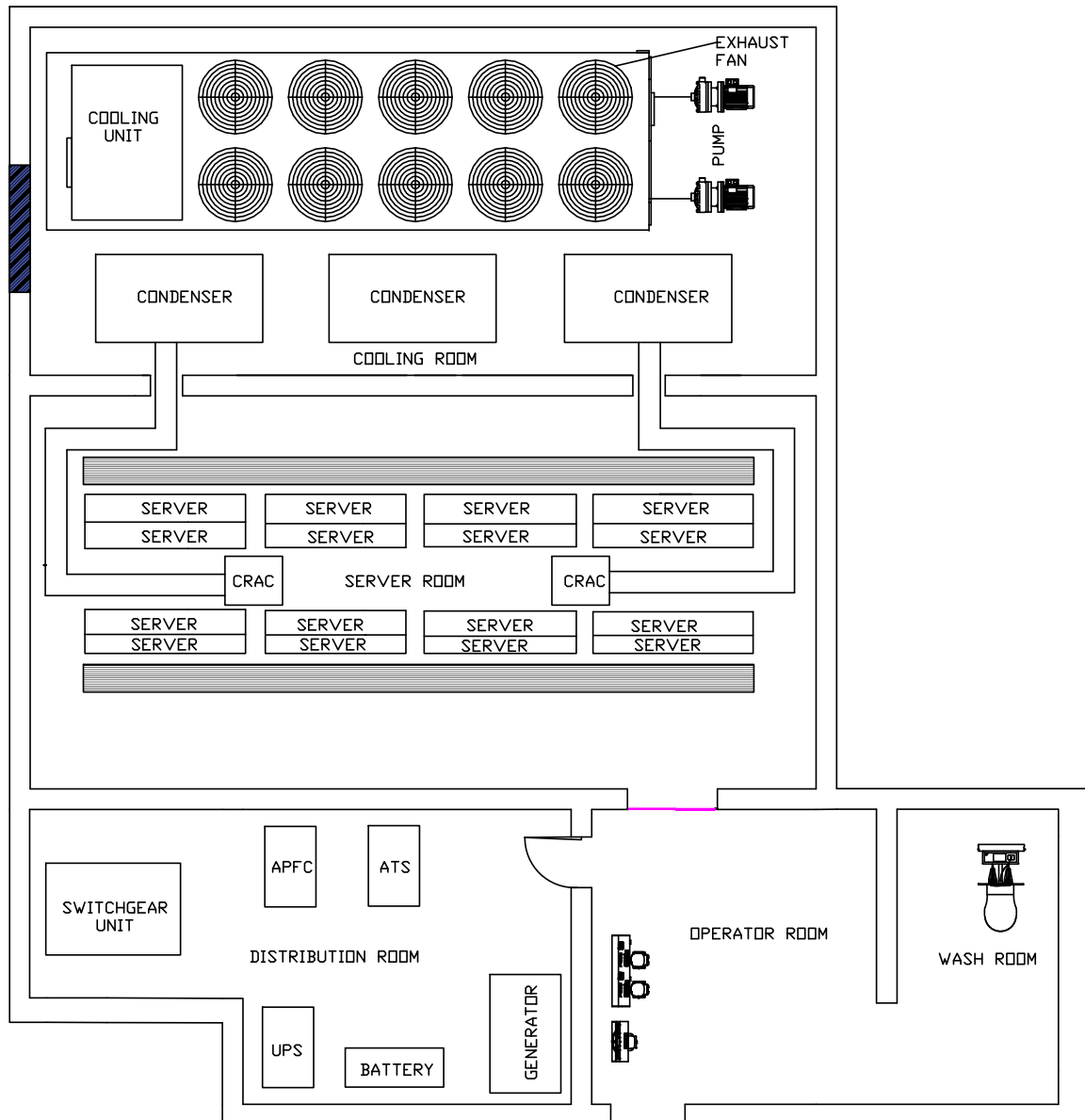


Figure 2: Layout Plan

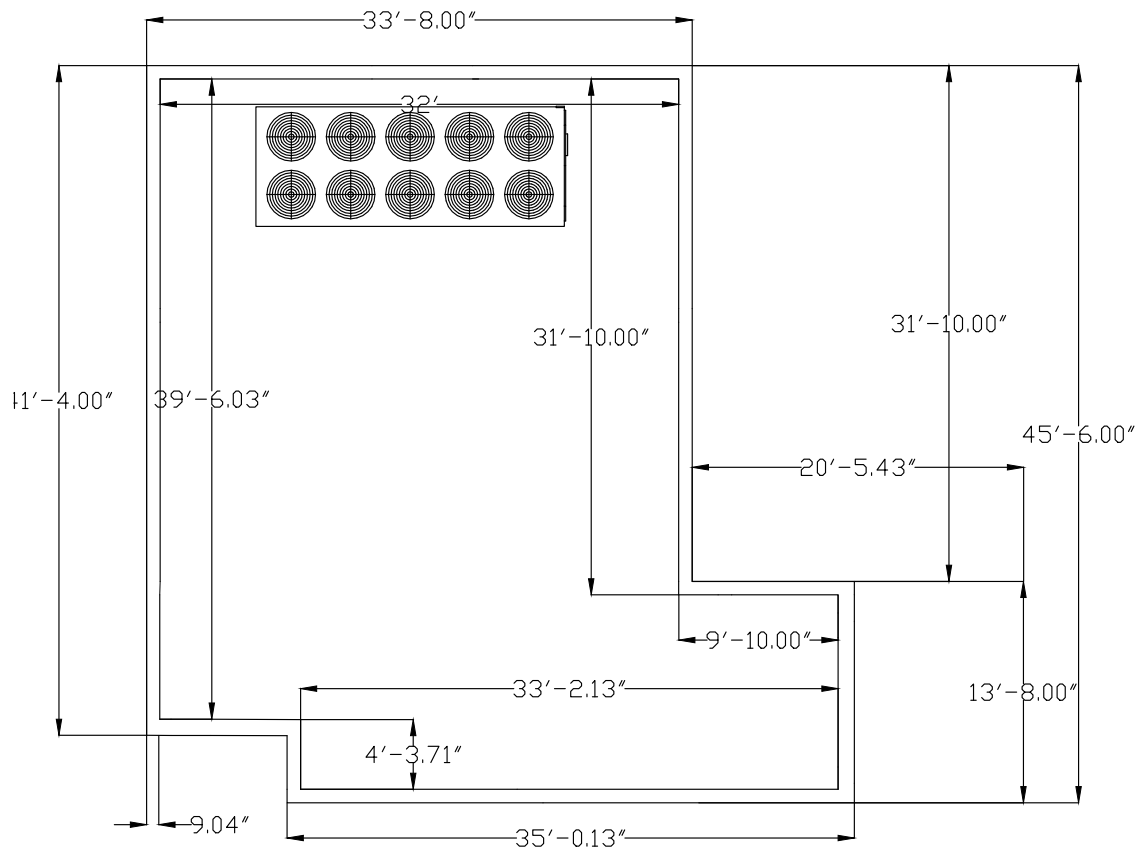


Figure 3: Rooftop Layout Measurement

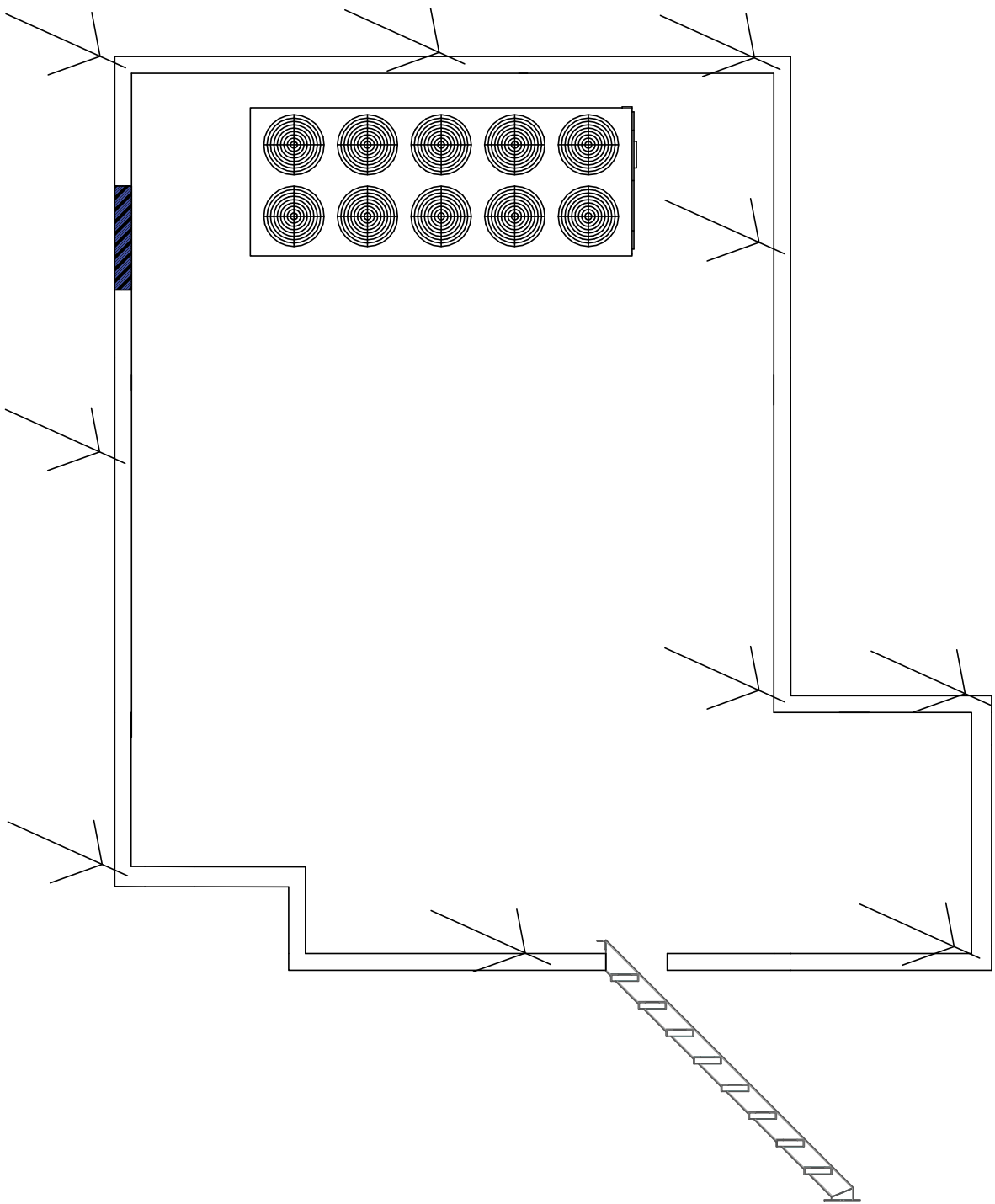


Figure 4: Rooftop Planning

3 Fitting and Fixture Details

ELECTRICAL SYMBOL	
	MAIN DISTRIBUTION BOARD (MDB)
	EMERGENCY DISTRIBUTION BOARD (EMDB)
	SUB DISTRIBUTION BOARD (SDB)
	EMERGENCY SUB DISTRIBUTION BOARD (ESDB)
	SWITCH BOARD (SB)
	FAN (F)
	LINEAR LED (LL)
	LED PANEL (LP)
	CEILING LIGHT (CL)
	2PIN SOCKET (2PS)
	3PIN SOCKET (3PS)
	ISOLATOR SWITCH (RIS)

Figure 5: Equipment Block

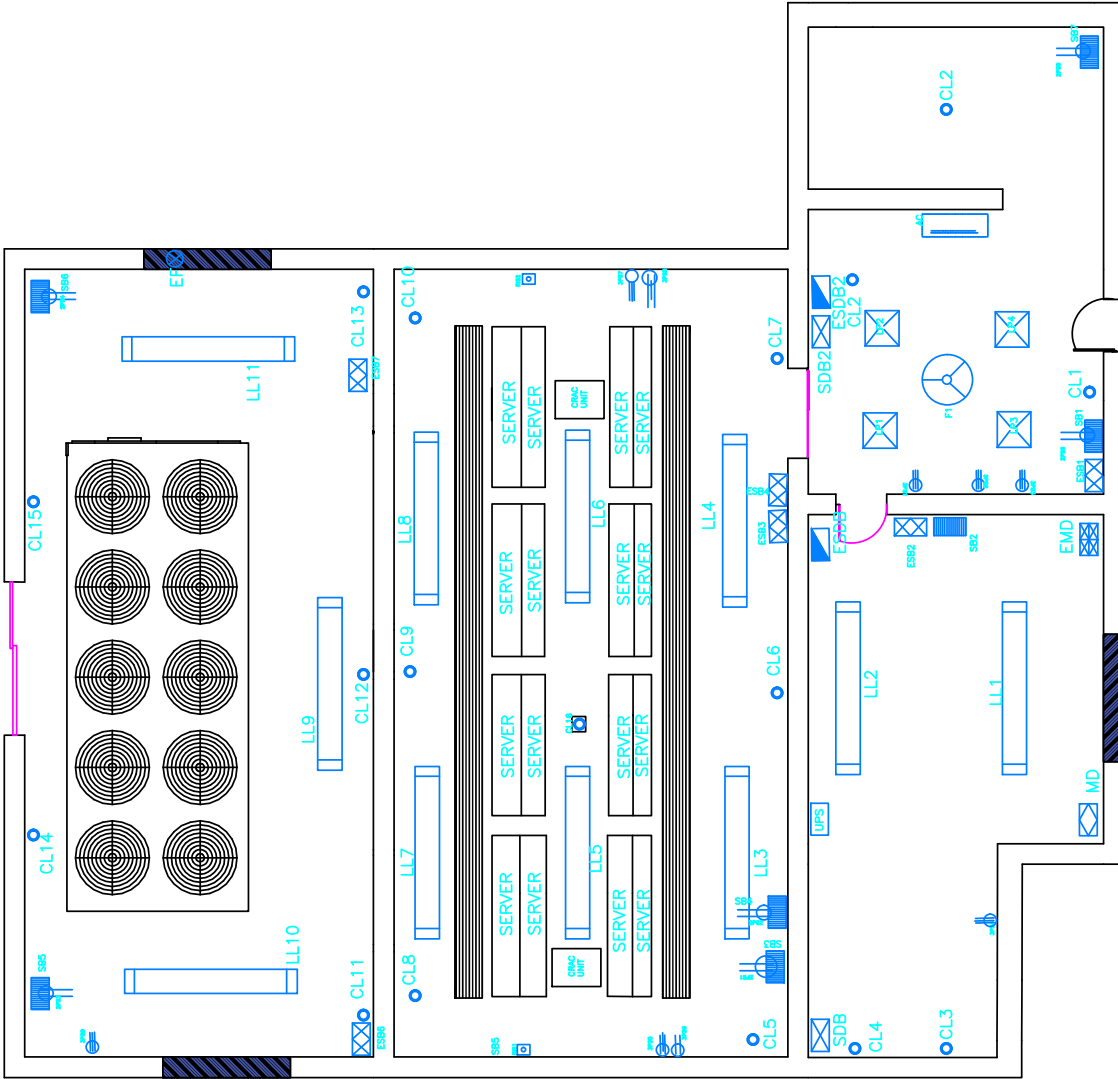


Figure 6: Fixtures and Fixtures

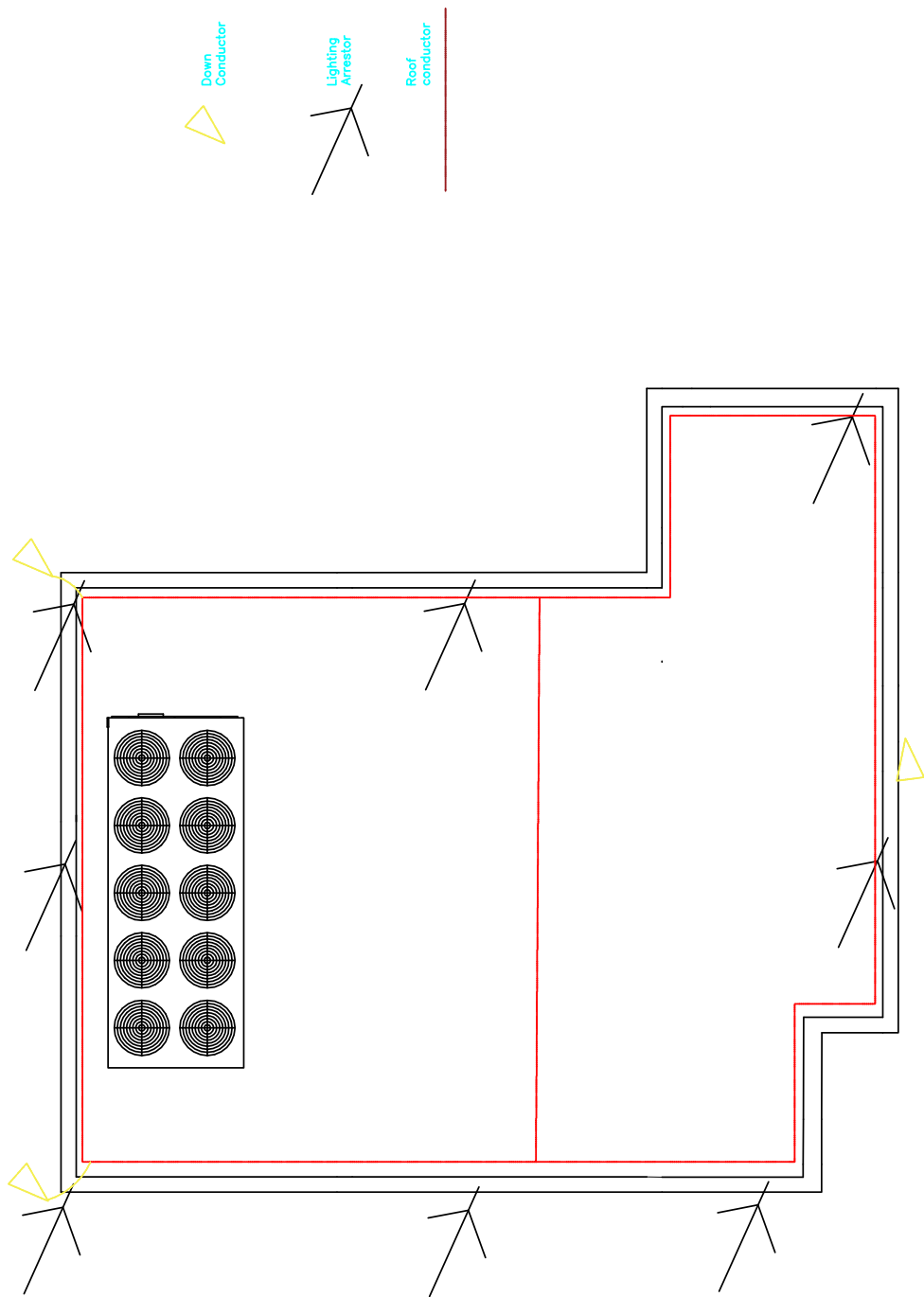


Figure 7: Roof Conduit Layout

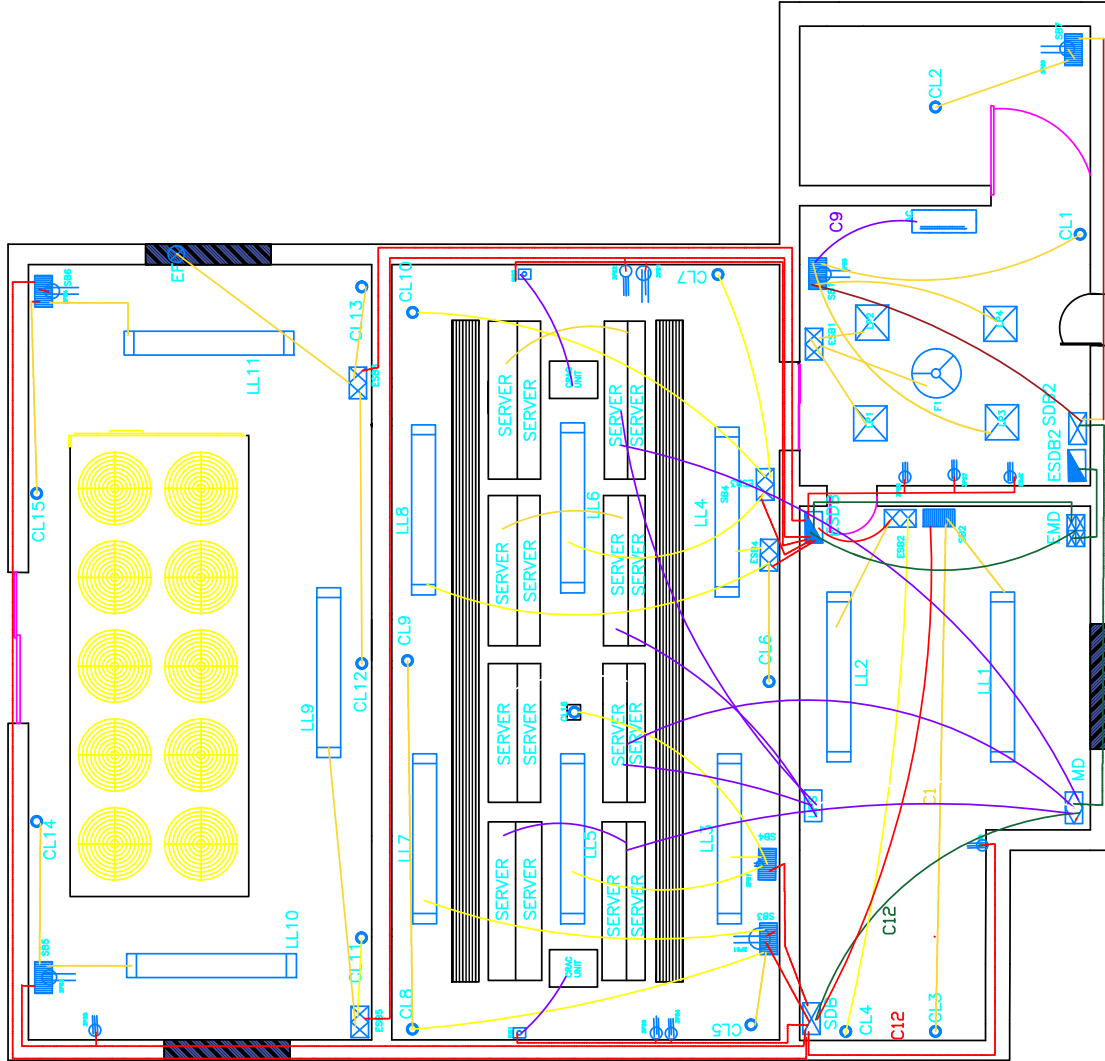


Figure 8: Final Conduit Layout

4 Single Line Diagrams (SLD)

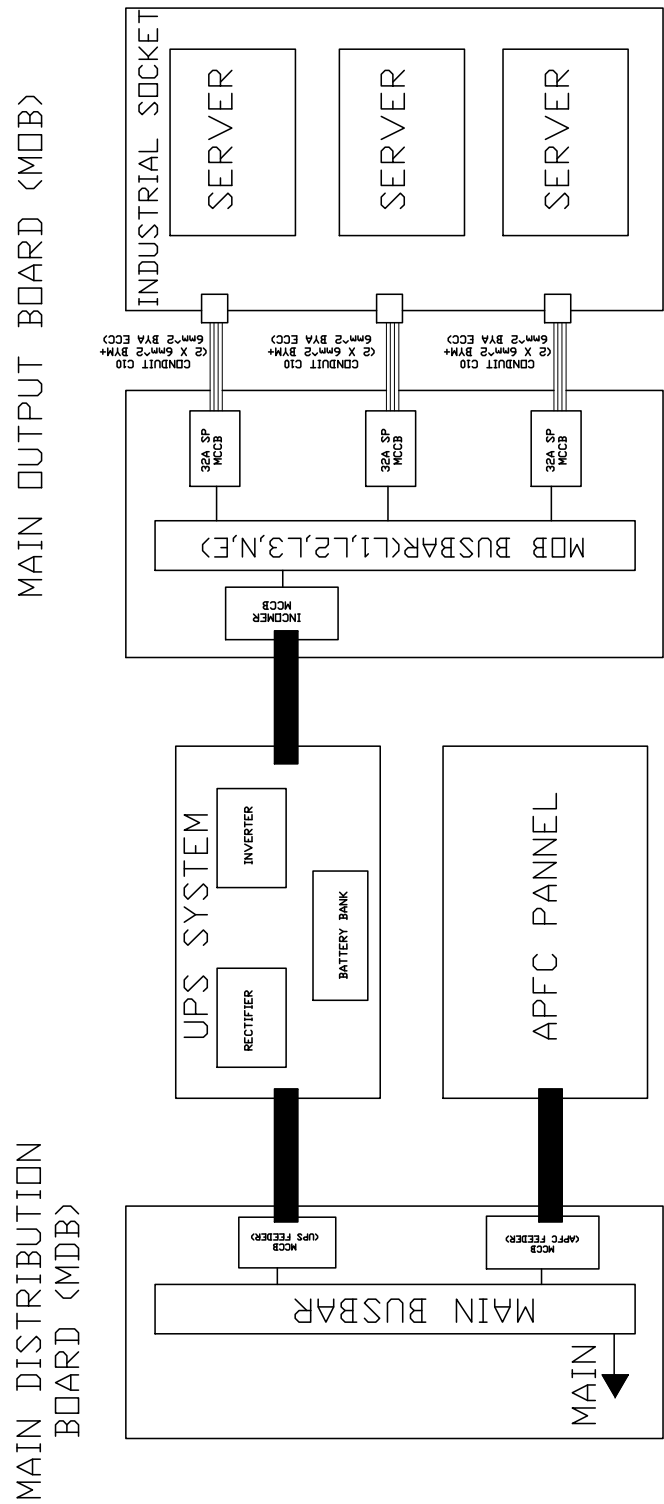


Figure 9: Main Distribution to Server SLD

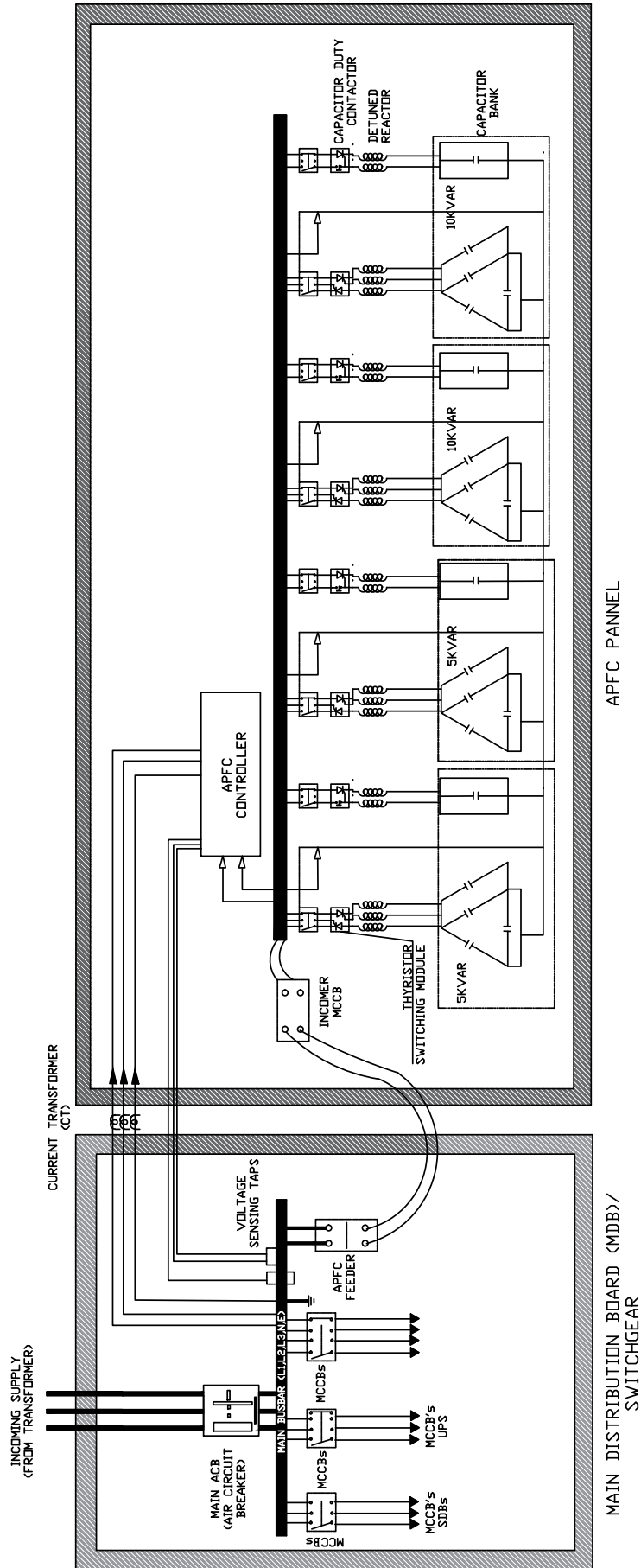


Figure 10: Automatic Power Factor Correction System SLD

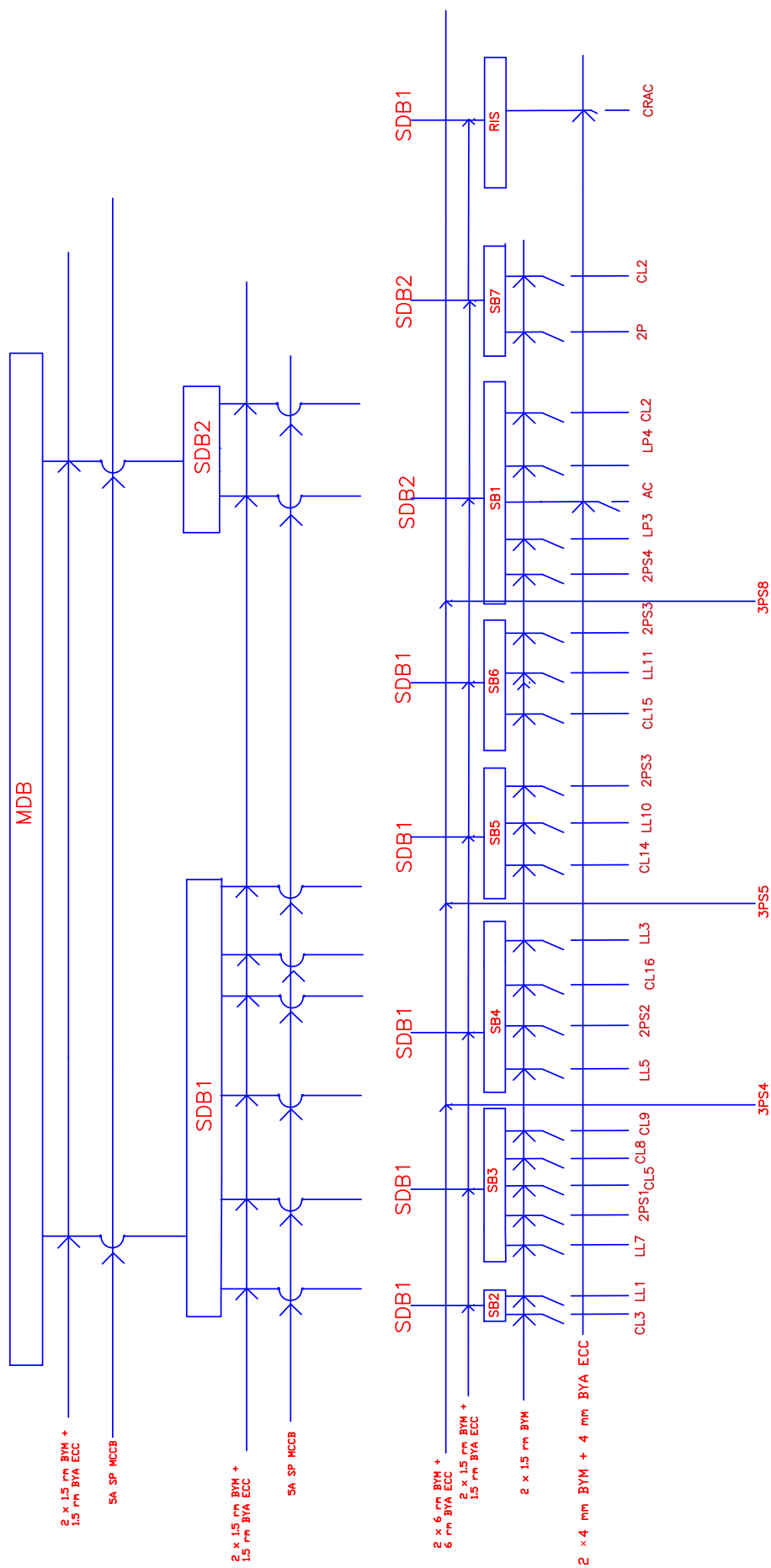


Figure 11: Sub-Distribution Board (Power) SLD

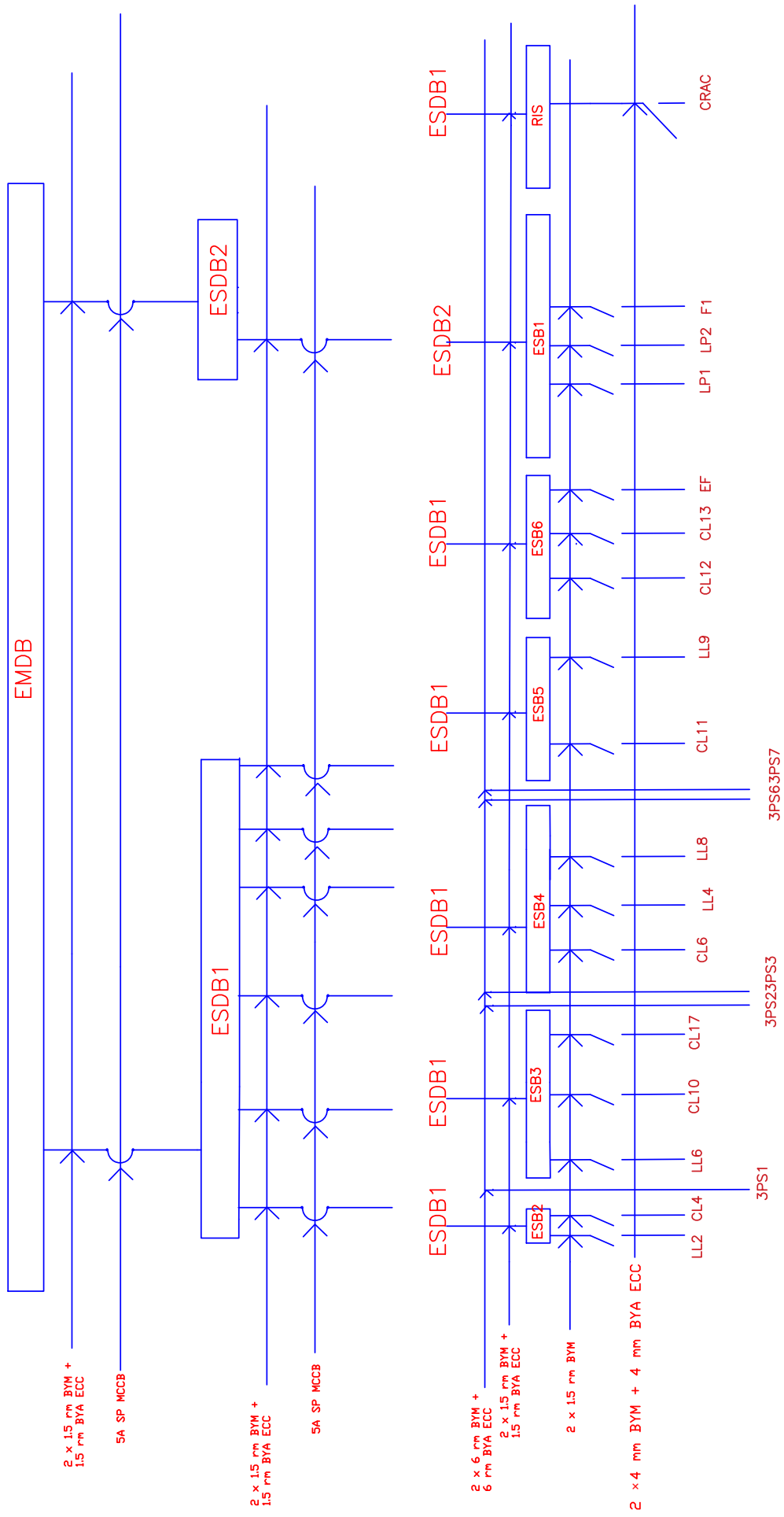


Figure 12: Emergency Sub-Distribution Board SLD

5 Lighting Design

6 Lighting Design Calculations

The luminance requirements for each technical room were computed using standard lighting engineering formulas. Below are the detailed calculations for the Operator Room, Cooling Room, Electrical Control Room, Washroom, and Server Room.

6.1 Operator Room

- **Room Dimensions:** Length = 11.60 ft, Width = 12.01 ft

$$\text{Area} = 11.60 \times 12.01 = 139.25 \text{ ft}^2 = 12.94 \text{ m}^2$$

- **Luminance Calculation:**

$$\text{Required Lumen} = \frac{12.94 \times 500}{0.6 \times 0.8} = 13,479.17 \text{ lumen}$$

- **Light Selection:** Four 30 W LED Panels ($2' \times 2'$) and one 30 W Ceiling Light:

$$\text{Lumen Output} = (5 \times 30) \times 100 = 15,000 \text{ lumen}$$

6.2 Cooling Room

- **Room Dimensions:** Length = 32.00 ft, Width = 14.17 ft

$$\text{Area} = 32.00 \times 14.17 = 453.33 \text{ ft}^2 = 42.12 \text{ m}^2$$

- **Luminance Calculation:**

$$\text{Required Lumen} = \frac{42.12 \times 200}{0.6 \times 0.8} = 17,550.00 \text{ lumen}$$

- **Light Selection:** Three 40 W Linear LEDs and two 30 W Ceiling Lights:

$$\text{Lumen Output} = (3 \times 40 + 2 \times 30) \times 100 = 18,000 \text{ lumen}$$

6.3 Electrical Control Room

- **Room Dimensions:** Combined Sections (L-Shape), Total Area = 21.09 m²

- **Luminance Calculation:**

$$\text{Required Lumen} = \frac{21.09 \times 300}{0.6 \times 0.8} = 13,181.25 \text{ lumen}$$

- **Light Selection:** Two 40 W Linear LEDs and two 30 W Ceiling Lights:

$$\text{Lumen Output} = (2 \times 40 + 2 \times 30) \times 100 = 14,000 \text{ lumen}$$

6.4 Washroom

- **Room Dimensions:** Length = 12.01 ft, Width = 6.57 ft, Area = 7.33 m²

- **Luminance Calculation:**

$$\text{Required Lumen} = \frac{7.33 \times 150}{0.6 \times 0.8} = 2,290.63 \text{ lumen}$$

- **Light Selection:** One 30 W Ceiling Light:

$$\text{Lumen Output} = 1 \times 30 \times 100 = 3,000 \text{ lumen}$$

6.5 Server Room

- **Room Dimensions:** Length = 32.00 ft, Width = 16.00 ft, Area = 47.57 m²
- **Luminance Calculation:**

$$\text{Required Lumen} = \frac{47.57 \times 500}{0.6 \times 0.8} = 49,552.08 \text{ lumen}$$

- **Light Selection:** Six 40 W Linear LEDs and six 30 W Ceiling Lights:

$$\text{Lumen Output} = (6 \times 40 + 6 \times 30) \times 100 = 42,000 \text{ lumen}$$

Lighting Summary Tables

Table 1: Room Luminance Requirements

Room Name	Area (m ²)	Target Lux	U.F.	M.F.	Required Lumen	Status
Operator Room	12.94	500	0.6	0.8	13,479.17	OK
Cooling Room	42.12	200	0.6	0.8	17,550.00	OK
Elec. Control Room	21.09	300	0.6	0.8	13,181.25	OK
Washroom	7.33	150	0.6	0.8	2,290.63	OK
Server Room	47.57	500	0.6	0.8	49,552.08	Acceptable

Table 2: Light Wattage Calculation (Efficacy: 100 lm/W)

Room Name	Req. Lumen	Qty 30W	Qty 40W	Total Watt	Calc. Lumen	Avg Lux
Operator Room	13,479	5	0	150	15,000	556
Cooling Room	17,550	2	3	180	18,000	205
Elec. Control Room	13,181	2	2	140	14,000	318
Washroom	2,290	1	0	30	3,000	196
Server Room	49,552	6	6	420	42,000	441

7 Load Calculation Analysis

7.1 Emergency Electrical Load & Circuit Analysis (ESDB)

Table 3: Emergency Switchboard Component & Load Schedule

SDB	SB	Fixture/Load	Qty	Unit(W)	Total(W)	Amps(A)	Cond.	Wire Rating	Brkr
ESDB1	ESB2	LL2 (Linear LED)	1	40	70	0.37	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		CL4 (Ceiling Light)	1	30					
ESDB1	ESB3	LL6, CL10, CL17	3	Mixed	100	0.53	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		(1x40W + 2x30W)	-	-					
		3PS1 (Power Socket)	1	500	500	2.67	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	15A
ESDB1	ESB4	CL6, LL4, LL8	3	Mixed	110	0.59	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		(1x30W + 2x40W)	-	-					
		3PS2, 3PS3	2	500	1000	5.35	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	15A
ESDB1	ESB5	CL11 (Ceiling Light)	1	30	70	0.37	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		LL9 (Linear LED)	1	40					
		3PS6, 3PS7	2	500	1000	5.35	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	15A
ESDB1	ESB6	CL12, CL13	2	30	160	0.86	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		EF (Exhaust Fan)	1	100					
EDB2	ESB1	LP1, LP2 (Panel)	2	30	140	0.75	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		F1 (Fan)	1	80					
ESDB1	RIS	CRAC Unit	1	4000	4000	21.39	C9	2x6.0 rm BYM + 6.0 rm BYA ECC	30A

7.2 Normal Electrical Load & Circuit Analysis (SDB)

Table 4: Switchboard Component & Load Schedule

SDB	SB	Fixture/Load	Qty	Unit(W)	Total(W)	Amps(A)	Cond.	Wire Rating	Brkr
SDB1	SB2	CL3 (Ceiling Light)	1	30	70	0.37	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		LL1 (Linear LED)	1	40					
SDB1	SB3	LL7 (Linear LED)	1	40	130	0.69	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		CL5, CL8, CL9	3	30					
		2PS1 (Socket)	1	100	600	3.21	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	10A
		3PS4 (Power Socket)	1	500					
SDB1	SB4	LL5, LL3	2	40	110	0.59	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		CL16	1	30					
		2PS2 (Socket)	1	100	100	0.53	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	10A
SDB1	SB5	LL10	1	40	70	0.37	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		CL14	1	30					
		2PS3, 3PS5	2	600	600	3.21	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	10A
SDB1	SB6	LL11	1	40	70	0.37	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		CL15	1	30					
		2PS3, 3PS8	2	600	600	3.21	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	10A
SDB2	SB1	LP3, LP4 (Panel)	2	30	90	0.48	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		CL2	1	30					
		2PS4	1	100	100	0.53	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	10A
		AC (Air Conditioner)	1	2000	2000	10.70	C9	2x4.0 rm BYM + 4.0 rm BYA ECC	20A
SDB2	SB7	CL2 (Corridor)	1	30	30	0.16	C1	2x1.5 rm BYM + 1.5 rm BYA ECC	5A
		2P (Socket)	1	100	100	0.53	C8	2x2.5 rm BYM + 2.5 rm BYA ECC	10A
SDB2	RIS	CRAC Unit	1	4000	4000	21.39	C9	2x6.0 rm BYM + 6.0 rm BYA ECC	30A

8 Facility Load Summation

The total electrical load is categorized into Normal (SDB) and Emergency (ESDB) paths based on the facility's lighting and power schedules.

8.1 Normal Sub-Distribution Board (SDB) Loads

The normal load summation is derived from the standard switchboard schedule [?].

Table 5: Normal SDB Load Breakdown

Board	Components	Detail	Load (W)
SDB 1	SB2, SB3, SB4, SB5, SB6	Lighting & Sockets	2,350
SDB 2	SB1, SB7	Panel, AC, Sockets	2,320
SDB 2	RIS	CRAC Unit 1	4,000
Total Normal Load (P_{SDB})			8,670

8.2 Emergency Sub-Distribution Board (ESDB) Loads

The emergency load summation includes critical infrastructure supported by the backup generator .

Table 6: Emergency ESDB Load Breakdown

Board	Components	Detail	Load (W)
ESDB 1	ESB2–ESB6	Emergency Lighting & Power	3,010
ESDB 1	RIS	CRAC Unit 2	4,000
EDB 2	ESB1	Panel & Fan	140
Total Emergency Load (P_{ESDB})			7,150

8.3 Critical & Mechanical Loads

- **Critical UPS Load:** For 16 Server Racks at 2 kW per rack, inclusive of 10% charging overhead:

$$P_{UPS} = (16 \times 2,000 \text{ W}) \times 1.1 = \mathbf{35,200 \text{ W}}$$

- **Mechanical Load:** Two water pumps for facility cooling/safety:

$$P_{Pump} = 2 \times 4,000 \text{ W} = \mathbf{8,000 \text{ W}}$$

9 Detailed Component Calculations

9.1 Pump Motor Sizing (Per Unit)

- **Power Rating (P):** 4,000 W
- **Operating Current (I):**

$$I = \frac{P}{\sqrt{3} \times V_L \times PF} = \frac{4000}{\sqrt{3} \times 381 \times 0.8} = \frac{4000}{527.79} = \mathbf{7.57 \text{ A}}$$

- **Protection:** 10 A TP MCCB
- **Cable Specification:** $2 \times 2.5 \text{ rm BYM} + 2.5 \text{ rm BYA ECC}$

9.2 Emergency Main Distribution Board (EMDB) Sizing

The EMDB supplies power to all critical loads (ESDB, UPS, Pumps) backed by the generator.

- **Total Connected Load (P_{Conn}):**

$$P_{Conn} = P_{ESDB} + P_{UPS} + P_{Pump} = 7,150 + 35,200 + 8,000 = \mathbf{50,350\text{ W}}$$

- **Design Active Load:** Applying a Demand Factor (DF) of 0.7:

$$P_{EMDB} = 50,350 \times 0.7 = \mathbf{35,245\text{ W}}$$

- **Rated Current:**

$$I_{EMDB} = \frac{35,245}{527.79} = \mathbf{66.78\text{ A}}$$

- **Selection:** 80 A 4-Pole MCCB

9.3 Generator Sizing

The generator must support the active EMDB load during utility failure.

- **Required Active Power:** 35.25 kW
- **Apparent Power Rating (kVA):** Assuming $PF = 0.8$:

$$S_{Gen} = \frac{35.25}{0.8} = \mathbf{44.06\text{ kVA}}$$

- **Recommended Unit:** 60 kVA
- **Justification:** A 60 kVA generator provides adequate operational headroom ($\approx 73\%$ load), adhering to Tier 3 standards for reliability and future capacity growth.

9.4 Main Distribution Board (MDB) Sizing

The MDB integrates the Normal (SDB) and Emergency (EMDB) paths.

- **Total MDB Load:**

$$P_{MDB} = (P_{SDB} \times 0.7) + P_{EMDB} = (8,670 \times 0.7) + 35,245$$

$$P_{MDB} = 6,069 + 35,245 = \mathbf{41,314\text{ W}}$$

- **Rated Current:**

$$I_{MDB} = \frac{41,314}{527.79} = \mathbf{78.28\text{ A}}$$

- **Selection:** 125 A 4-Pole MCCB
- **Justification:** While a 100 A breaker is theoretically sufficient, a **125 A** breaker is selected to align with standard transformer outputs and provide a safety margin for inrush currents.

10 Transformer and Power Quality Design

10.1 Transformer Rating

Based on the facility's main current draw:

- **Voltage Ratio:** 11 kV / 415 V
- **Calculated Capacity:**

$$S_{Trans} = \frac{\sqrt{3} \times 415 \times 78.28}{1000} \approx \mathbf{56.26 \text{ kVA}}$$

- **Recommended Unit:** 100 kVA
- **Feasibility:** A 100 kVA transformer operates at approximately 56% capacity. This is optimal for data centers to handle the non-linear loads of IT equipment and allows for significant future expansion.

10.2 Automatic Power Factor Correction (APFC)

To improve the Power Factor from 0.8 to a target of 0.99 for the 41.31 kW load.

- **Required Reactive Power (Q_C):**

$$Q_C = P \times [\tan(\cos^{-1} \phi_1) - \tan(\cos^{-1} \phi_2)]$$

$$Q_C = 41.31 \times (0.75 - 0.142) = 41.31 \times 0.608 = \mathbf{25.12 \text{ kVAR}}$$

- **Recommended Bank:** 30 kVAR Panel
- **Configuration:** Distributed in steps (5 + 5 + 10 + 10 kVAR) to ensure precise correction and switching redundancy.

11 Bill of Quantity (BOQ) & Cost Analysis

This section provides the consolidated financial estimation for the electrical infrastructure of the Tier 3 Data Center. The pricing is based on local premium manufacturers (Energypac, Bizli, Super Star) to ensure reliability.

11.1 Critical Power Infrastructure

Table 7: Critical Power Infrastructure Costs

Item Description	Specification	Qty	Rate (Tk)	Total (Tk)
Diesel Generator	60 kVA (Energypac)	1	1,250,000	1,250,000
Transformer	100 kVA (11/0.415 kV)	1	480,000	480,000
Precision Cooling (CRAC)	4 kW (Stulz/Vertiv)	2	450,000	900,000
Online UPS	45 kVA (N+1 Config)	1	850,000	850,000
Subtotal				3,480,000

11.2 Power Distribution & Quality Panels

Table 8: Distribution & APFC Panel Costs

Item Description	Specification	Qty	Rate (Tk)	Total (Tk)
MDB Panel	125A 4P Main	1	95,000	95,000
EMDB Panel	80A 4P Main	1	65,000	65,000
APFC Panel	30 kVAR (Detuned)	1	145,000	145,000
Subtotal				305,000

11.3 Cost Breakdown: 30 kVAR Detuned APFC Panel

Table 9: APFC Panel Internal Breakdown

Component Category	Technical Detail	Cost (Tk)
Detuned Reactors	7% Impedance, Copper Wound	42,000
Thyristor Modules	4-Step High-Speed Switching	35,000
Capacitor Banks	Heavy Duty Delta-Connected (Total 30 kVAR)	18,500
Enclosure & Busbars	Powder-coated Industrial Cabinet	18,500
Protection Gear	Incomer MCCB, HRC Fuses, CTs	16,500
APFC Controller	Intelligent Microprocessor Relay	14,000
Total Panel Cost	Fully Assembled & Tested	145,000

11.4 Lighting, Utilities & RIS Integration

Table 10: Utilities & Fixture Costs

Item Description	Brand / Model	Qty	Rate (Tk)	Total (Tk)
RIS Dist. Panel	Custom 30A (For CRAC)	2	45,000	90,000
Linear LED (40W)	Super Star (4 ft)	11	1,450	15,950
LED Panel (30W)	Super Star (2' × 2')	4	2,850	11,400
Ceiling Light (30W)	Super Star PSA-30W	16	700	11,200
Exhaust Fan (EF)	Vision (12 inch)	1	3,250	3,250
Ceiling Fan (F1)	Vision (56 inch)	1	3,250	3,250
Switchboards	Superstar (Mixed Gang)	14	296	4,144
3-Pin Socket (15A)	Super Star	8	450	3,600
2-Pin Socket (10A)	Super Star	6	350	2,100
Subtotal				144,894

11.5 Wiring & Containment

Table 11: Cabling & Containment Costs

Material	Specification	Length (m)	Rate (Tk)	Total (Tk)
Main Feeder Cable	6.0 rm BYM (Bizli)	150	285.00	42,750
Socket Circuit	2.5 rm BYM (Bizli)	250	145.50	36,375
Lighting Circuit	1.5 rm BYM (Bizli)	450	104.40	46,980
Earth (ECC)	1.5/2.5 rm BYA	700	37.03	25,921
PVC Conduit	Vigo Guard (3/4")	550	95.00	52,250
Subtotal				204,276

11.6 Grand Total Summary

Table 12: Final Project Cost Summary

Project Segment	Cost (Tk)
1. Critical Power Infrastructure	3,480,000
2. Power Distribution (MDB/EMDB/APFC)	305,000
3. Lighting, Utilities & RIS Panels	144,894
4. Wiring & Containment	204,276
Total Estimated Cost	4,134,170

12 Conclusion

This report successfully outlines the comprehensive electrical system design for a Tier-3 Data Center, integrating an Automatic Power Factor Correction (APFC) unit to ensure compliance with Bangladesh energy standards and improve operational efficiency.

Key deliverables achieved in this design include:

- A lighting design that meets the 500 Lux standard for server rooms while maintaining high energy efficiency (approx. 100 lm/W).
- A robust load calculation that separates Critical Loads (Servers, CRAC) from Utility Loads (Lights, Fans), ensuring appropriate breaker sizing.
- A detailed cost analysis that accounts for top-tier redundancy (N+1 UPS) and safety features.

By implementing the detuned APFC panel and the dedicated RIS distribution network, the proposed system is designed to provide 99.982% availability, minimizing downtime and ensuring a stable environment for critical IT infrastructure.